



Bayesian Approach to Hurst Exponent Estimation

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Abstract Fractal investigation of a signal often involves estimating its fractal dimension or Hurst exponent H when considered as a sample of a fractional process. Fractional Gaussian noise (fGn) belongs to the family of self-similar fractional processes and it is dependent on parameter H . There are variety of traditional methods for Hurst exponent estimation. Our novel approach is based on zero-crossing principle and signal segmentation. Thanks to the Bayesian analysis, we present a new axiomatically based procedure of determining the expected value of Hurst exponent together with its standard deviation and credible intervals. The statistical characteristics are calculated at the interval level at first and then they are used for the deduction of the aggregate estimate. The methodology is subsequently used for the EEG signal analysis of patients suffering from Alzheimer disease.

Keywords Fractal dimension · Hurst exponent · Bayesian approach · EEG · Alzheimer disease

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1 Introduction

Fractal character of a signal is often investigated via estimation of its Hurst exponent when considered as a sample of fractional process. Fractional Brownian motion (fBm) and fractional Gaussian noise (fGn) are believed to be the ones of the most important processes with fractional character. The fractal assumption of data is used in many fields of research - in network traffic modelling (Ming 2010), financial time series (Masaaki and Tam 2013) or biomedicine (Maxim et al. 2005). According to the theory, the fBm process can be converted to fGn with the same Hurst exponent by means of differencing, and similarly we can obtain fBm sample in the opposite direction.

Precise Hurst exponent estimation is useful, because it often carries more information than autocorrelation. Despite both characteristics describe the dependency of the investigated signal, the Hurst exponent provides also information about self-similarity and memory of the original process. Another alternative to this exponent is fractal dimension, which is often used for general fractal structures without fBm assumption.

Traditional methods of Hurst exponent estimation utilize important properties of the fractional processes. Rescaled range method (R/S) (Mandelbrot and Wallis 1969) belongs to the well-known statistical procedures of fractal dimension estimation in time domain, however due to its low accuracy it can be used in fields where excessive precision is not required such as geography (Bogusz et al. 2016) or traffic flow modelling (Xue et al. 2015). In the frequency domain one can use the Whittle estimator (Whittle 1953) utilizing the shape of power spectrum of a signal. With respect to Holder continuity of fBm, the Quadratic variations method (Istas and Lang 1997) can provide accurate estimation of Hurst exponent and the traditional zero-crossing method estimates the parameter by means of counting the number of level crossings. Recently, there were certain attempts to refine the conventional methods using the self-affine property (Campanharo 2016), spectrum of a signal (Kristoufek 2014) or self-similarity (Fernandez-Martinez et al. 2014).

It is possible to test these methods on simulated data and determine the most accurate one. Results show that the zero-crossing and Quadratic variations are stable and precise methods. However, the main disadvantage of all traditional approaches is paradoxically the point estimation of Hurst exponent. When dealing with real data, it is possible to obtain only one number which represents the whole input.

Therefore, in this work we decided to upgrade the traditional zero-crossing approach and suggested a method of calculation of expected value of Hurst exponent together with its variance and credible intervals. This approach provides more information about the fractal relevance of investigated data and thanks to the segmentation employs more features calculated at the interval level.

2 Proposition of Zero Crossing

The number of zero-crossings (or level crossings) of Gaussian processes and its relevant fractal properties were published in several papers by Azais (1990), Feuerverger et al. (1994), and Couerjolly (2000). The main result of these studies is the dependence of Hurst parameter on the probability of zero-crossing of random continuous nowhere differentiable process at fixed interval. Our aim is to improve the method by analysing the number of zero-crossings of fGn process and to deduce the formula another way.

Suppose $x_k = x_k(H)$ is a sampling of fGn process with parameter H , zero mean and unit variance on fixed interval for $k = 1, \dots, N + 1$. The appropriate autocorrelation function ρ_k (Mandelbrot and Ness 1968) equals

$$\rho_k = \frac{1}{2} \left(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H} \right). \quad (1)$$

We denote the first autocorrelation coefficient by $\rho \equiv \rho_1 = 2^{2H-1} - 1$. The probability density function of two consecutive sample elements is

$$f(x_k, x_{k+1}) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp \left(-\frac{1}{2(1-\rho^2)} (x_k^2 - 2\rho x_k x_{k+1} + x_{k+1}^2) \right), \quad (2)$$

therefore the probability of zero-crossing equals

$$p^* = \text{prob}(x_k x_{k+1} < 0) \quad (3)$$

that can be expressed (Dlask et al. 2015) as

$$p^* = \frac{1}{\pi\sqrt{1-\rho^2}} \int_0^{+\infty} \int_0^{+\infty} \exp \left(-\frac{1}{2(1-\rho^2)} (x^2 + 2\rho xy + y^2) \right) dx dy. \quad (4)$$

Using the substitution $\xi = (\rho x + y)(1 - \rho^2)^{-\frac{1}{2}}$ and $\eta = x$ one obtains

$$p^* = \frac{1}{\pi} \iint_{O^*} \exp \left(-\frac{\xi^2 + \eta^2}{2} \right) d\xi d\eta \quad (5)$$

where $O^* = \left\{ (\rho x + y), x\sqrt{1-\rho^2} \right\} \in \mathbf{R}^2 : x, y \in \mathbf{R}_0^+ \}$. Substituting $\xi = r \cos \varphi$ and $\eta = r \sin \varphi$ the crossing probability can be expressed as

$$p^* = \frac{1}{\pi} \int_{\varphi_{\min}}^{\varphi_{\max}} d\varphi \cdot \int_0^{+\infty} \exp \left(-\frac{r^2}{2} \right) dr = \frac{\varphi_{\max} - \varphi_{\min}}{\pi}. \quad (6)$$

Due to the form of area O^* we have $\varphi_{\min} = 0$ and $\varphi_{\max}$ equals

$$\varphi_{\max} = \begin{cases} \arctan \frac{\sqrt{1-\rho^2}}{\rho} & \rho > 0 \\ \pi - \arctan \frac{\sqrt{1-\rho^2}}{|\rho|} & \text{otherwise} \end{cases}. \quad (7)$$

Utilizing Eqs. 6 and 7, it is possible to express the probability p^* as

$$p^* = \frac{1}{\pi} \operatorname{arccot} \frac{\rho}{\sqrt{1-\rho^2}} \quad (8)$$

and finally using the identity $\rho = 2^{2H-1} - 1$, the point estimate of Hurst parameter H equals

$$H = 1 + \log_2 \cos \frac{\pi p^*}{2}, \quad (9)$$

which is revisited form of well-known formula published by Couerjolly (2000), originally in the form

$$H = \frac{1}{2} \left(1 + \log_2 (1 + \epsilon |\cos(\pi \cdot S_N)|) \right), \quad (10)$$

where the relative number of zero crossing S_N with respect to the sample length N and $\epsilon = \text{sgn}\left(H - \frac{1}{2}\right)$ are assumed to be known.

3 Bayesian Approach

We already know how to obtain point estimate of H with given fGn sample of length $N + 1$ – if the number of zero-crossings in the sample equals Z , we can approximately express H as

$$H \approx 1 + \log_2 \cos \frac{\pi Z}{2N}. \quad (11)$$

Unfortunately, from formula (11) we do not know the distribution of Hurst exponent H neither the distribution of zero-crossings p^* . Suppose, that the probability p^* is known for investigated sample. Then the number of zero-crossings Z comes from binomial distribution $Z \sim \text{Bi}(N, p^*)$ with probability density function

$$f(Z|p^*) = \binom{N}{Z} (p^*)^Z \cdot (1 - p^*)^{N-Z}. \quad (12)$$

Using the Bayesian inverse rule (Gelman 2003), the posterior probability density equals

$$f(p^*|Z) = \frac{(p^*)^Z (1 - p^*)^{N-Z}}{B(N + 1, N - Z + 1)}, \quad (13)$$

where $B(p, q)$ is the standard beta function. We can generalize the result by means of using Beta prior as

$$f_{\text{PRIOR}}(p^*) = \frac{(p^*)^{-a} (1 - p^*)^{-a}}{B(1 - a, 1 - a)}, \quad (14)$$

for parameter $a \in [0, 1)$. After the application of Bayesian rule (Dlask et al. 2015), we get the posterior probability for p^* in the form

$$f_{\text{POST}}(p^*|Z) = \frac{(p^*)^{Z-a} (1 - p^*)^{N-Z-a}}{B(Z + 1 - a, N - Z + 1 - a)}, \quad (15)$$

therefore the probability p^* comes from Beta distribution $p^* \sim B(Z + 1 - a, N - Z + 1 - a)$. Different estimates of p^* can be obtained for different choices of parameter a , however it is reasonable to consider only natural Bayesian approach with $a = 0$ or Jeffreys-Perks law (Jeffreys 1945; Perks 1947) with $a = 1/2$.

4 Segmentation of the Signal

When trying to find different fractal patterns (different H values) in the investigated signal, it is necessary to divide the original sampling into disjoint segments and determine the statistical features in each interval separately. Function (15) describes the probability of zero-crossing in the fGn sample in general – considering the segmentation into L disjoint segments, each with $N + 1$ elements, the probability $f_L(p)$ of zero-crossing of the whole signal sample with $L \cdot (N + 1)$ elements equals

$$f_L(p) = \frac{1}{L} \sum_{k=1}^L \frac{p^{Z_k-a} (1 - p)^{N-a-Z_k}}{B(Z_k + 1 - a, N - Z_k + 1 - a)}, \quad (16)$$

where Z_k is the number of zero-crossings in k -th segment. Based on the shape of the aggregate probability density $f_L(p)$ one can calculate the mean value of Hurst parameter as

$$E(H) = 1 + \int_0^1 f_L(p) \log_2 \cos \frac{p\pi}{2} dp \quad (17)$$

and variance

$$\sigma^2 = \int_0^1 \left(1 + \log_2 \cos \frac{p\pi}{2}\right)^2 f_L(p) dp - (E(H))^2. \quad (18)$$

with fixed number of signal divisions L .

As easy to prove, fGn of length N has the same mean value and variance of H estimate for any nonoverlapping segmentation of constant segment length M , where $M < N$. But in the case of real signal, the fGn assumption with constant Hurst parameter is not always fulfilled and the number of zero-crossings in each segment can be different, therefore the mean and variance can vary.

5 Method of Estimation

Using different segmentations of the signal, it is possible to determine the crucial statistical characteristics that are dependent on the parameter L . In this section we would like to present the key idea of the whole procedure by means of showing the optimal number of non-overlapping subintervals L^* . To formalize such an idea, it is necessary to introduce two axioms:

1. unimodality of the $f_L(p)$ function,
2. minimization of the number of segments L , where $L > 1$.

The second requirement can be replaced by a stricter condition, that the variance of Hurst exponent estimation has to be minimized. However, considering real data, these two conditions often coincide.

For certain L^* large enough the aggregate probability density function becomes unimodal and we denote this number as the *optimal segmentation* of the given sample defined as

$$L^* = \begin{cases} 1 & \text{for } f_L(p) \text{ is unimodal for all allowable } L \\ \min \{L > 1 : f_L(p) \text{ is unimodal}\} & \text{otherwise} \end{cases}$$

On the basis of the function $f_L^*(p)$ we can determine the boundaries $p_{\min}$, $p_{\max}$ of credible interval for p with confidence level $\tilde{\alpha}$ as

$$\int_0^{p_{\min}} f_L^*(p) dp = \frac{\tilde{\alpha}}{2}, \quad (19)$$

$$\int_0^{p_{\max}} f_L^*(p) dp = 1 - \frac{\tilde{\alpha}}{2}. \quad (20)$$

Finally, we obtain the boundaries of credible interval for H as

$$H_{\min} = 1 + \log_2 \cos \frac{p_{\max}\pi}{2}, \quad (21)$$

$$H_{\max} = 1 + \log_2 \cos \frac{p_{\min}\pi}{2}, \quad (22)$$

where $p_{\min}$ and $p_{\max}$ were calculated from (19) and (20) with given confidence level $\tilde{\alpha}$.

In the traditional frequentist method one could calculate the credible intervals by means of utilizing the properties of standard binomial distribution. Under the assumption that the number of zero-crossings $Z \sim \text{Bi}(N, p^*)$ denote that

$$\text{prob}(Z = k) = p_k = \binom{N}{k} (p^*)^k (1 - p^*)^{N-k}. \quad (23)$$

One determines the highest value of non-negative integer A and the lowest value of non-negative integer B satisfying conditions

$$\sum_{k=0}^A p_k \leq \alpha/2, \quad (24)$$

$$\sum_{k=B}^N p_k \leq \alpha/2, \quad (25)$$

where α is the confidence level. The adequate boundaries of credible interval for Hurst exponent equal

$$H_{\min}^{ZC} = 1 + \log_2 \cos \frac{B\pi}{2N}, \quad (26)$$

and

$$H_{\max}^{ZC} = 1 + \log_2 \cos \frac{A\pi}{2N}. \quad (27)$$

6 Case Study - Application to EEG

Methodology of optimal segmentation can be subsequently used for analysis of electroencephalographic signal (EEG). The aim is to find different patterns in two types of signal and distinguish between healthy patients (CN) and those who suffer from Alzheimer disease (AD). The subject of the analysis is 146 EEG samples of CN and 28 samples of AD patients.

Human brain anatomy and function display self-similarities at different spatial and temporal scales and a combination of modular and hierarchical organization (Lohse et al. 2014). The fractal dimension of EEG is optimally sensitive to the neuronal dysfunction secondary to a brain lesion (Zappasodi et al. 2014). Different EEG techniques have been used to explore differences between normal controls and early AD with varying success. So development of novel biomarkers of AD is necessary (Antila et al. 2013).

EEG signal is considered to be quasi-stationary in each two second interval. Therefore it is possible to use the estimation method for any window with 400 elements assuming the sampling rate is 200 Hz. Because the EEG signal time series is significantly longer, it is possible to investigate the Hurst exponent estimate in every quasi-stationary interval, that means in every non-overlapping window with 400 elements. Thanks to this approach, we obtain the dependence of Hurst exponent on segmentation index and can investigate its behaviour in time. The development of Hurst exponent in EEG sample of a diseased patient is visualised in Fig. 1, whereas the Fig. 2 shows the dependence in the case of CN.

Based on the estimations of expected value of Hurst exponent and corresponding credible interval boundaries it is possible to utilize different measures of variability to find difference between healthy and diseased patients. Range (R), interquartile range (IQR), standard deviation (s), median absolute deviation (MAD) and absolute average deviation (AAD) were examined if they carry the crucial information that statistically distinguishes

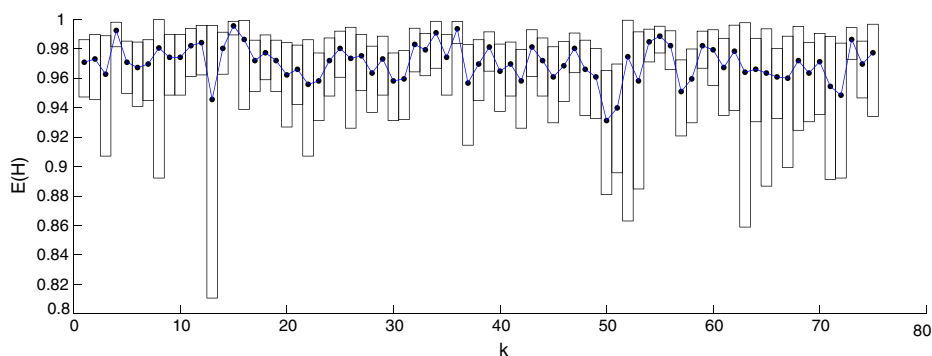


Fig. 1 Time variability of Hurst exponent in case of Alzheimer disease

the two patient groups. Due to the non-normality of output data, we use non-parametric Wilcoxon-Mann-Whitney test (WMW) to find out whether the two samples are from the same population.

From all considered measure variabilities, range of Hurst exponents was proven to be the most efficient in EEG analysis. Figure 3 shows the range fluctuation in case of second channel. The expected values of range in case of diseased (E_{AD}) and normal (E_{CN}) patients together with their standard deviation s and the p -value from WMW are presented in Table 1.

The lowest p -values were obtained in the case of channel 7,17 and 18. Using the false discovery rate principle (Hochberg and Benjamini 1990) by means of multiple testing on significance level $\alpha = 0.05$ we performed $19 \times 5 = 95$ WMW tests of null hypothesis H_0 indicating median equality in both groups. Due to the repeated testing, we must diminish critical probability to $\alpha_{FDR} = 0.0158$. Significant cases are highlighted in bold in Table 2.

In the case of range as fluctuation measure, we observed significant differences between AD and CN on channels: 1-4,6-7,9,11-13 and 17-19. Concerning IQR as a measure of variability, the substantial differences appeared on channels 3,13 and 17-19. In the case of standard deviation s the lowest p -values were observed on channels 3,13,12-13 and 17-19. As far as MAD is concerned, according to FDR principle, the difference appeared only on

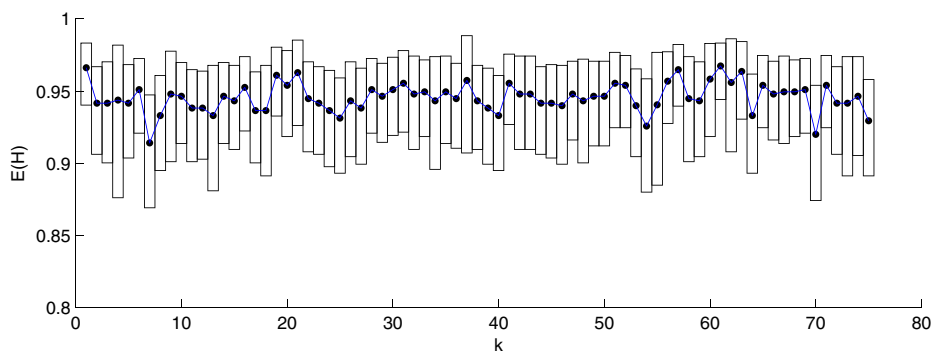


Fig. 2 Time variability of Hurst exponent in case of Control normal

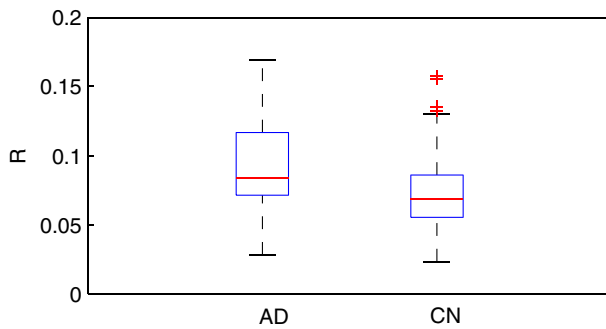


Fig. 3 Fluctuation of range in case of seventh channel

channels 3,13 and 17–19. The AAD measure was not useful for distinguishing between AD and CN patients. The electrode scheme of EEG is visualised in Fig. 4.

On the basis of calculated p -values, the range of Hurst exponent value was identified as the most significant characteristic to distinguish between AD and CN patients also in the case of traditional methods. Table 3 shows the differences in range as p -value for all channels and the novel method (Bayesian zero-crossing, BZC), traditional zero-crossing method (TZC) and the R/S method. The critical probability was $\alpha_{\text{FDR}} = 0.0017$ in the case of TZC and $\alpha_{\text{FDR}} = 0.0006$ using R/S. Using the FDR principle again and traditional zero-crossing method, there were only four channels where we rejected the null hypothesis about equal means and in the case of R/S, the difference was recorded only on channels 7

Table 1 Expected values and standard deviations of range mean values of Hurst exponent

channel	E_{AD}	s_{AD}	E_{CN}	s_{CN}	p val.
1	0.0943	0.0328	0.0886	0.1261	0.0070
2	0.0915	0.0335	0.0867	0.1299	0.0064
3	0.1066	0.0453	0.0907	0.1315	0.0006
4	0.0952	0.0387	0.0895	0.1313	0.0081
5	0.0879	0.0361	0.0870	0.1333	0.0189
6	0.0929	0.0387	0.0870	0.1319	0.0049
7	0.1074	0.0408	0.0908	0.1313	0.0001
8	0.1091	0.0598	0.0941	0.1298	0.0341
9	0.0958	0.0426	0.0894	0.1321	0.0158
10	0.0893	0.0418	0.0838	0.1301	0.0402
11	0.0928	0.0374	0.0870	0.1313	0.0147
12	0.1089	0.0485	0.0943	0.1305	0.0037
13	0.1034	0.0508	0.0880	0.1307	0.0022
14	0.0935	0.0450	0.0937	0.1782	0.0174
15	0.0873	0.0409	0.0809	0.1295	0.0334
16	0.0876	0.0396	0.0823	0.1294	0.0229
17	0.1036	0.0414	0.0856	0.1296	0.0001
18	0.1053	0.0457	0.0859	0.1302	0.0003
19	0.0996	0.0381	0.0871	0.1299	0.0009

Table 2 Significance of differences between AD and CN as p -value

channel	R	IQR	s	MAD	AAD
1	0.0070	0.0295	0.0193	0.0445	0.9984
2	0.0064	0.1153	0.0468	0.1832	0.4139
3	0.0006	0.0009	0.0005	0.0010	0.0441
4	0.0081	0.1144	0.0472	0.1812	0.5403
5	0.0189	0.2769	0.0758	0.4046	0.6245
6	0.0049	0.0571	0.0520	0.0638	0.6657
7	0.0001	0.0301	0.0050	0.0486	0.2842
8	0.0341	0.2278	0.1172	0.2842	0.8170
9	0.0158	0.3069	0.0725	0.4377	0.9201
10	0.0402	0.1785	0.0738	0.1963	0.9136
11	0.0147	0.0650	0.0428	0.0668	0.6776
12	0.0037	0.0520	0.0121	0.0693	0.6362
13	0.0022	0.0049	0.0056	0.0042	0.1344
14	0.0174	0.0550	0.0535	0.0621	0.6657
15	0.0334	0.3600	0.0858	0.4723	0.9396
16	0.0229	0.1035	0.0295	0.1102	0.9624
17	0.0001	0.0004	0.0001	0.0009	0.1144
18	0.0003	0.0007	0.0003	0.0010	0.1070
19	0.0009	0.0058	0.0005	0.0077	0.2169

and 17. The previous results demonstrate the advantage of signal segmentation – whereas the traditional method consider the signal as a whole and are trying to find the most suitable aggregate Hurst exponent value, the novel method adapts to the partial Hurst exponents in the optimal segmentation and determines its expected value based on the average of partial probability densities.

Fig. 4 Electrode scheme 10-20

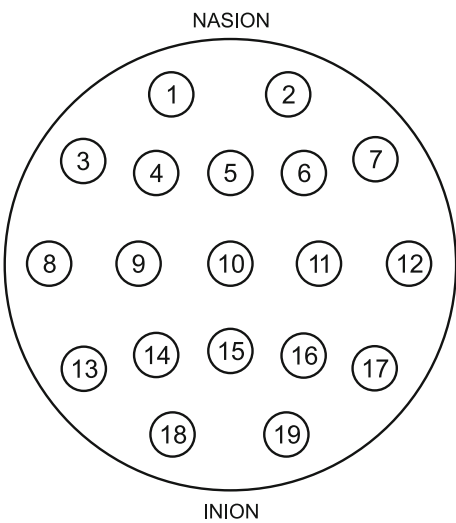


Table 3 Comparison of new and traditional method as a difference in range

channel	BZC	TZC	R/S
1	0.0070	0.0186	0.0606
2	0.0064	0.0641	0.0289
3	0.0006	0.0130	0.0443
4	0.0081	0.0240	0.1431
5	0.0189	0.1632	0.2174
6	0.0049	0.0481	0.0918
7	0.0001	0.0005	0.0003
8	0.0341	0.2550	0.1794
9	0.0158	0.1328	0.2845
10	0.0402	0.0803	0.1351
11	0.0147	0.0083	0.0375
12	0.0037	0.0197	0.1266
13	0.0022	0.0010	0.0081
14	0.0174	0.0231	0.0952
15	0.0334	0.0356	0.1578
16	0.0229	0.0805	0.0653
17	0.0001	0.0014	0.0006
18	0.0003	0.0017	0.0031
19	0.0009	0.0418	0.0083

7 Conclusion

Bayesian approach together with zero-crossing principle was proven to be a suitable tool for fractal analysis especially for estimating the Hurst exponent of time series. Thanks to the segmentation of quasi-stationary signal we were able to consider only stationary intervals for the calculation of aggregate estimate. Fluctuation of expected value of Hurst exponent over segments was used for classification of AD and CN patients and the most significant differences were observed on channels 3,13 and 17-19.

Studies have shown less complexity of the EEG signals, and more perturbations in EEG synchrony (Jeong 2004). Higher variability of Hurst exponent in AD patients in this study may be related to more perturbations in EEG synchrony. Loss of neurons but also perturbed anatomical and/or functional coupling may result in impairments in the temporal coordination of distributed neuronal activity, resulting in higher variability of Hurst exponent. The most significant changes were found in the brain areas with the most expressed neurohistological changes (frontotemporal regions). The distribution of changes are similar to those described for changes in complex noise characteristics (Vysata et al. 2014).

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References

- Antila K, Lötjönen J, Thurfjell L, Laine J, Massimini M et al. (2013) The PredictAD project: development of novel biomarkers and analysis software for early diagnosis of the Alzheimer's disease. *Interface Focus* 3. doi:[10.1098/rsfs.2012.0072](https://doi.org/10.1098/rsfs.2012.0072)
- Azais JM (1990) Conditions for convergence of number of crossings to the local time: Application to stable processes with independent increments and to Gaussian processes. *Probab Math Stat* 11:19–36
- Bogusz J, Klos A, Figurski M, Kujawa M (2016) Investigation of long-range dependencies in the stochastic part of daily GPS solutions. *Surv Rev* 48:140–147
- Campanharo A (2016) Hurst exponent estimation of self-affine time series using quantile graphs. *Physica A - Statistical Mechanics and its Applications* 444:43–48
- Couerjolly JF (2000) Simulation and identification of the fractional Brownian motion: a bibliographical and comparative study. *J Stat Softw* 5:1–53
- Couillard M, Davison M (2005) A comment on measuring the Hurst exponent of financial time series. *Phys A* 348:404–418
- Glask M, Kukal J, Tran VQ (2015) Revisited Zero-Crossing Method for Hurst Exponent Estimation in Time Series. *MME Proceedings* 1:115–120
- Fernandez-Martinez M, Sanchez-Granero MA et al. (2014) An accurate algorithm to calculate the Hurst exponent of self-similar processes. *Phys Lett A* 378:2355–2362
- Feuerverger A, Hall P, Wood A (1994) Estimation of Fractal Index and Fractal Dimension of Gaussian Process by Counting the Number of Level Crossings. *J Time Ser Anal* 15:587–606
- Gelman A (2003) *Bayesian Data Analysis*. 675 p. Chapman and Hall/CRC
- Hochberg Y, Benjamini Y (1990) More powerful procedures for multiple significance testing. *Stat Med* 9:811–818
- Hurst H (1956) Methods of Using Long-Term Storage in Reservoirs. *ICE Proceedings* 5:519–543
- Istas J, Lang P (1997) Quadratic variations and estimation of the local Holder index of a Gaussian process. *Ann Inst Henri Poincaré Probab Stat* 33:407–436
- Jeffreys H (1945) An invariant form for the prior probability in estimation problems. *Proc Roy Soc Lond A* 186:453–461
- Jeong J (2004) EEG Dynamics in patients with Alzheimer's disease. *Clin Neurophysiol* 115:1490–1505
- Kristoufek L (2014) Spectrum-based estimators of the bivariate Hurst exponent. *Phys Rev E* 90:1–6
- Lohse C, Bassett DS, Lim KO, Carlson JM (2014) Resolving Anatomical and Functional Structure in Human Brain Organization: Identifying Mesoscale Organization in Weighted Network Representations. *PLoS Comput Biol* 10(10):e1003712. doi:[10.1371/journal.pcbi.1003712](https://doi.org/10.1371/journal.pcbi.1003712)
- Mandelbrot B, Ness J (1968) Fractional Brownian motions, Fractional Noises and Applications. *SIAM Rev* 10:422–437
- Mandelbrot B, Wallis J (1969) Robustness of the rescaled range R/S in the measurement of noncyclic long run statistical dependence. *Water Resour Res* 5(5):967–988
- Masaaki K, Tam C (2013) Fractional Brownian Motions in Financial Models and Their Monte Carlo Simulation. *Theory and Applications of Monte Carlo Simulations* doi:[10.5772/53568](https://doi.org/10.5772/53568)
- Maxim V, Sendur L, Fadili J, Suckling J, Gould R, Howard R, Bullmore E (2005) Fractional Gaussian noise, functional MRI and Alzheimer's disease. *NeuroImage* 25:141–158
- Ming L (2010) Fractional Gaussian Noise and Network Traffic Modeling. *WSEAS International Conference on Applied Computer and Applied Computational Science* 1:34–39
- Perks W (1947) Some observations on inverse probability. *J Inst Actuar* 73:285–312
- Vysata O, Procházka A, Mares J, Rusina R, Pazdera L, Valis M, Kukal J (2014) Change in the characteristics of EEG color noise in Alzheimer's disease. *Clin EEG Neurosci* 45(3):147–151. doi:[10.1177/1550059413491558](https://doi.org/10.1177/1550059413491558)
- Sang H, Wang S (2001) Hurst exponent analysis of financial time series. *Journal of Shanghai University (English Edition)* 5:269–272
- Whittle P (1953) Estimation and information in stationary time series. *Ark Mat* 3:423–434
- Xue Y, Jia L, Teng W, Lu W (2015) Long-range correlations in vehicular traffic flow studied in the framework of Kerner's three-phase theory based on rescaled range analysis. *Communications in Non-linear Science and Numerical Simulation* 22:285–296
- Zappasodi F, Olejarczyk E, Marzetti L, Assenza G, Pizzella V, Tecchio F (2014) Fractal Dimension of EEG Activity Senses Neuronal Impairment in Acute Stroke. *PLoS One* 9(6):e100199. doi:[10.1371/journal.pone.0100199](https://doi.org/10.1371/journal.pone.0100199)